

ROBOT ARM



Introduction

The part of the project is on the arm control, crop recognition et crop picking. To do that, we will use the robot arm DofBot SE that we can find its specification on the annex file. Firstly, we will configure the environment of work, and after that we will read to see what the DofBot SE constructor gave us and how to use them. After that, we will attack the arm control.

Verification of the recognition of the robot arm by the computer:

-The stm32 card is a peripheric of the type of serial (UART).

Command to display it: `ls /dev/ttyUSB*`

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~$ ls /dev/ttyUSB* /dev/ttyACM*
ls: impossible d'accéder à '/dev/ttyACM*': Aucun fichier ou dossier de ce nom
/dev/ttyUSB0
```

The card is recognized as **ttyUSB0**.

- The camera is not a peripheric of serial communication, it sends a flow of heavy digital data.

Command to display it: `ls /dev/video*`

Before branching the camera:

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~$ ls /dev/video*
/dev/video0 /dev/video1
```

After branching the camera:

```
kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~$ ls /dev/video*
/dev/video0 /dev/video1 /dev/video2 /dev/video3
```

There are two names that appeared: **video2** and **video3**

This information will help us in the next to debug when some problem will occur.

Requirement

1- APP(IOS/Android):

- YahboomRobot.

It is the phone application for controlling the arm.

2- VM file

- VMware-workstation-full
- VM22.04_ROS2_DofBot-SE

They are the files needed to launch the virtual machine which contain the environment precisely made for the DofBot SE. There are the most important resources.

3- Instruction manual (no need for now)

It is a set of images to know how to manipulate the robot arm.

4- Hardware_3DModelFile

- Bus servo
- DOFBOT Hardware references Manual
- 3D Model file
- Stm32 board

They constitute the information about the hardware part.

5- PC_Software

- YahboomArmEn

This is the environment of simulation. It is a software made with the arm already spawned into it.

6- Code

- DofBotSE-Code-New

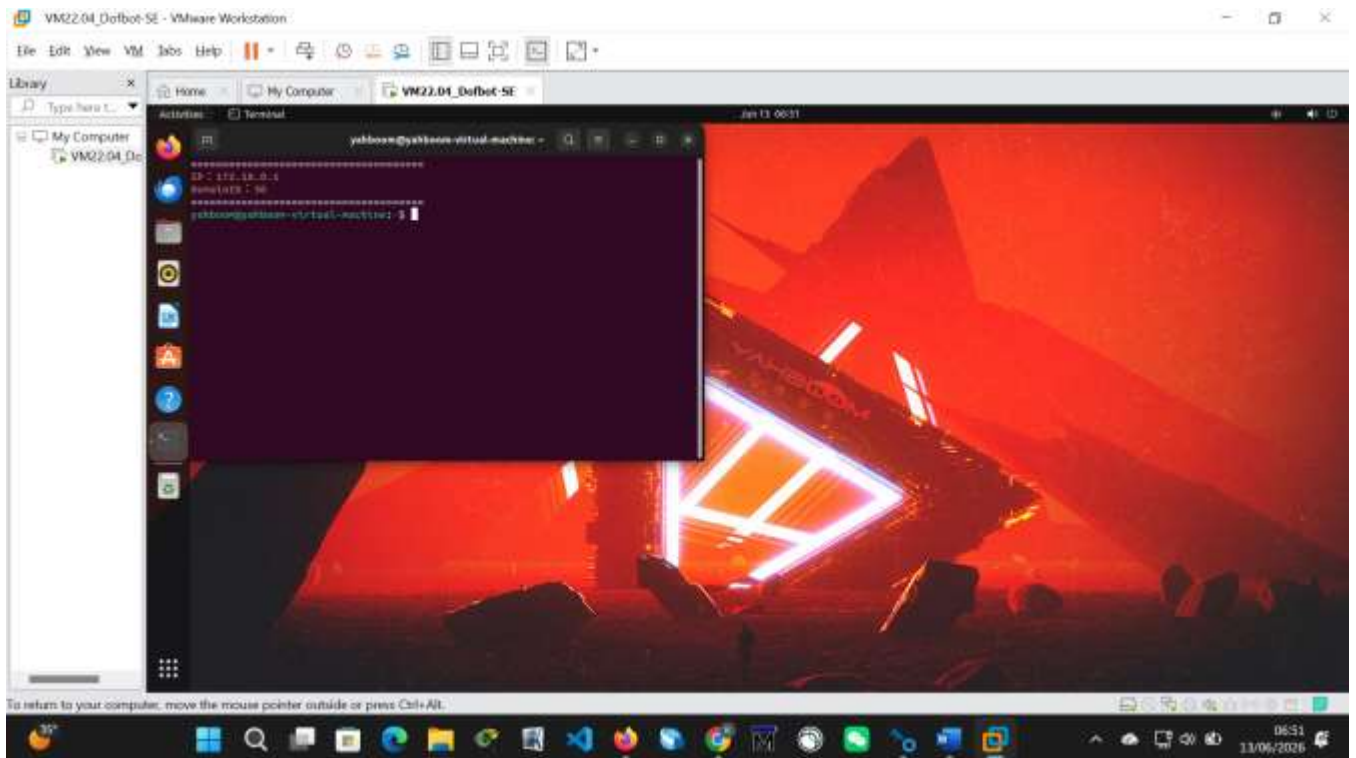
7- Communication Protocol-Python Library

- Install Python Module
- Stm32 and Virtual machine Serial port protocol
- MCU coprocessor I2C communication protocol
- PC APP remote control communication protocol

We did download everything the Yahboom website gave us, but the computer it is not enabled to extract the VM22.04_ROS2_DofBot-SE.zip because of a lack of storage. To deal with this problem, we will use an external storage SSD.

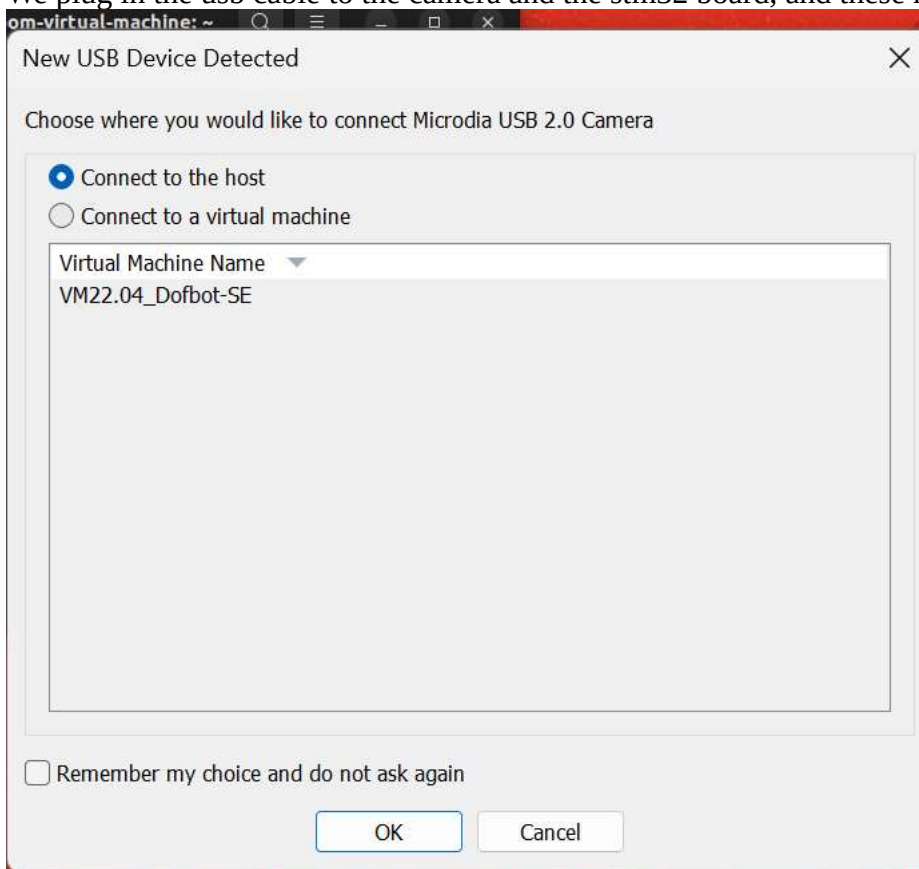
Problem of storage

After the installation of the software needed, we managed to get the virtual machine working.

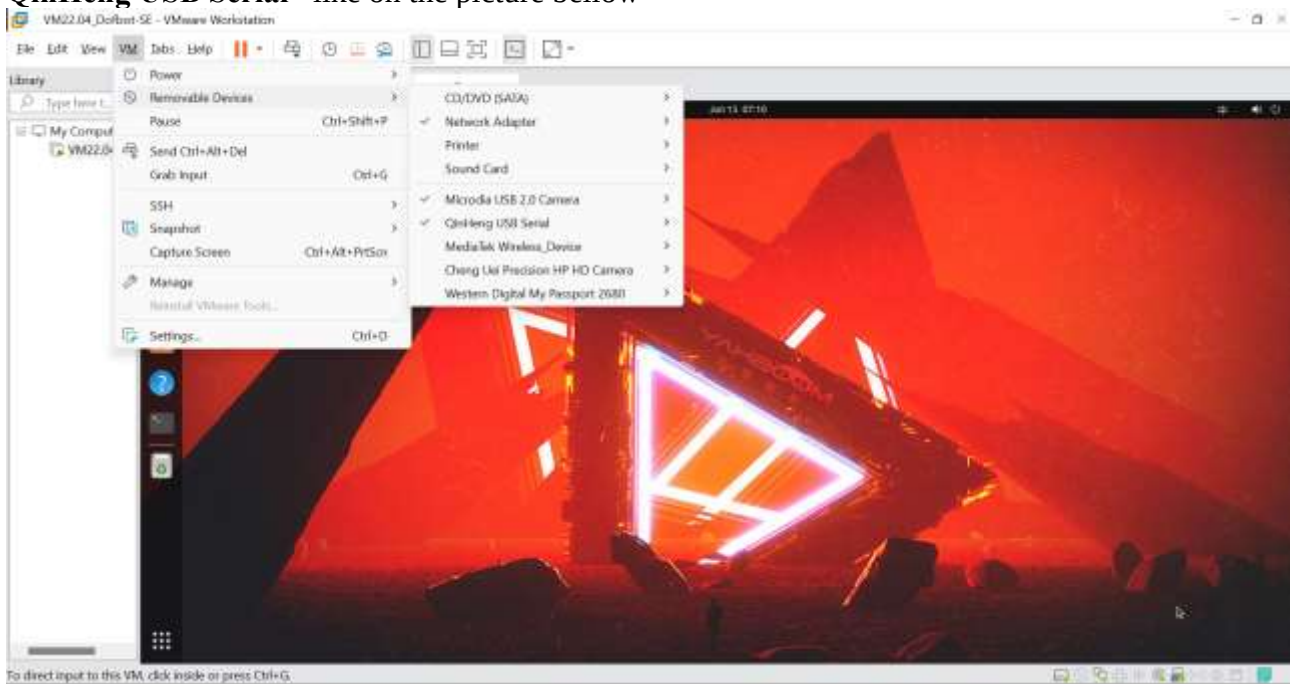


Configuration to connect the robot arm to the virtual machine

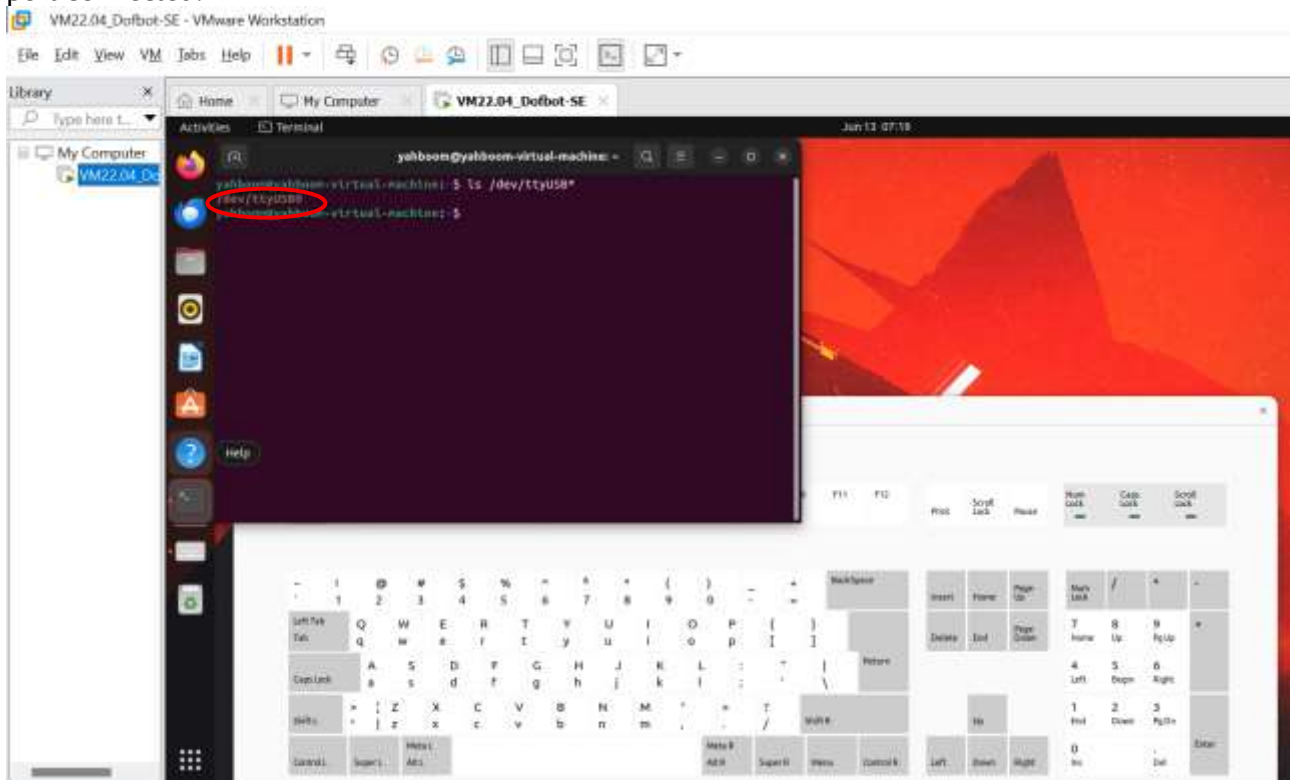
We plug in the usb cable to the camera and the stm32 board, and these messages will appear:



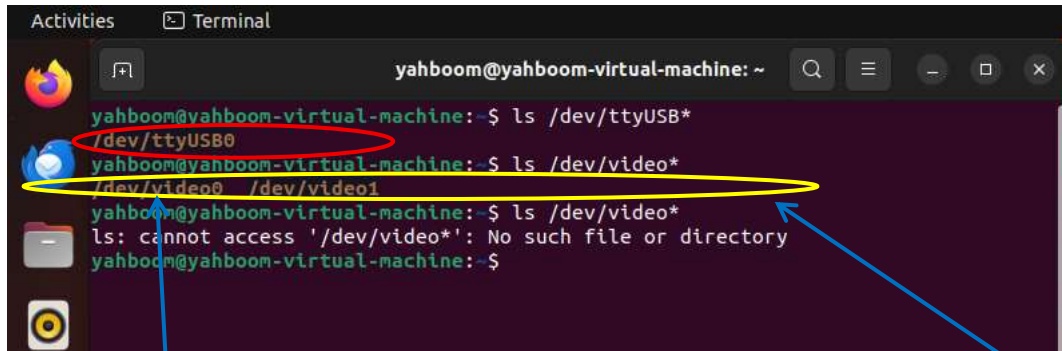
To connect the robot arm, choose “**connect to virtual machine** and select **VM22.04-DOFBOT-SE**” or directly do in the option **VM** and select “**network adapter, Microdi USB 2.0 Camera and QinHeng USB Serial**” like on the picture bellow



To verify if the robot arm and the camera are detected in the virtual machine, we can display the usb port connected:



For the camera:



The image shows a terminal window titled 'yahboom@yahboom-virtual-machine: ~'. The first command is `ls /dev/ttyUSB*`, which returns `/dev/ttyUSB0`. This line is circled in red. The second command is `ls /dev/video*`, which returns `/dev/video0 /dev/video1`. This line is circled in yellow. The third command is `ls /dev/video*`, which returns an error: `ls: cannot access '/dev/video*': No such file or directory`. Blue arrows point from the red and yellow circles to labels below the terminal.

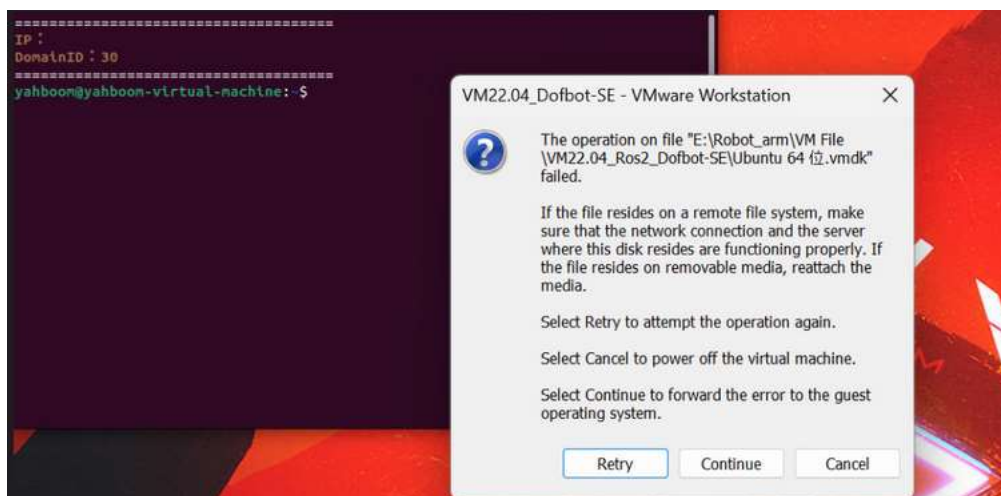
```
yahboom@yahboom-virtual-machine:~$ ls /dev/ttyUSB*
/dev/ttyUSB0
yahboom@yahboom-virtual-machine:~$ ls /dev/video*
/dev/video0 /dev/video1
yahboom@yahboom-virtual-machine:~$ ls /dev/video*
ls: cannot access '/dev/video*': No such file or directory
yahboom@yahboom-virtual-machine:~$
```

With the camera plugged

With the camera unplugged

Connexion to the robot arm

A problem appears when we try connecting the virtual machine workspace to the robot arm, here it is:

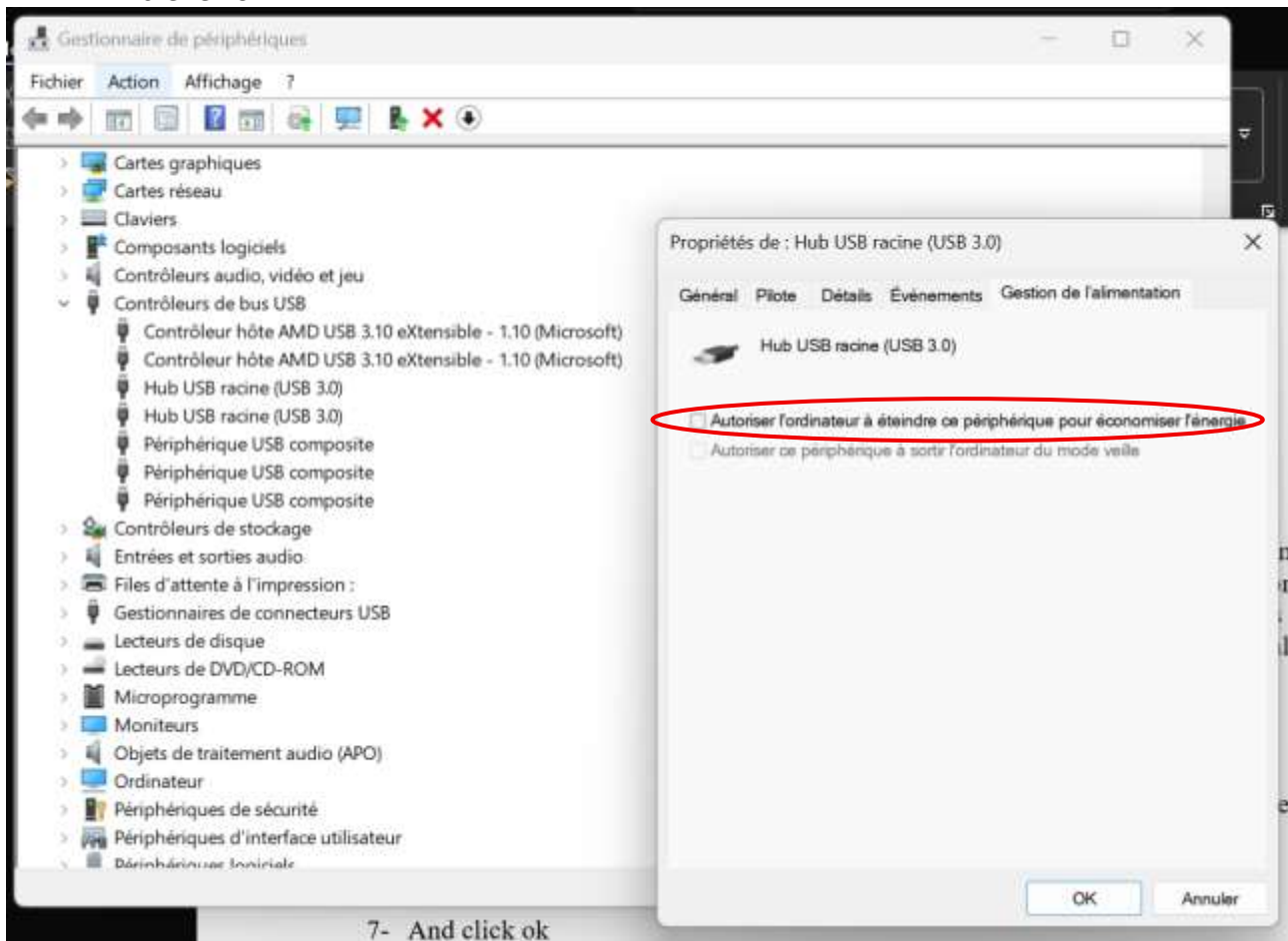


This problem is due to a windows IOS alimentation protection. When the VM is performing a resource-intensive task (such as loading ROS2 or running AI calculations), the drive consumes a spike in power. If your laptop's USB port doesn't provide enough power, or if Windows is trying to conserve energy, the drive “drops out” for a microsecond, abruptly cutting off the signal. To deal with this problem we must modify the Windows OS about the USB Hub.

To do so, we go in:

- 1- Device Manager.
- 2- Scroll down to the very bottom of the list until you see “USB Bus Controllers.”
- 3- Right-click on “USB Root Hub” (do this for each one if you have more than one)
- 4- Select Properties
- 5- Go to the Power Options tab
- 6- Uncheck the box: “Allow the computer to turn off this device to save power”

7- And click ok



7- And click ok

1- DOFBOT-JetsonNANO repository : https://www.yahboom.net/study/Dofbot-Jetson_nano

3- Tip :

<https://www.yahboom.net/public/upload/upload-html/1699442099/4.PC%20software%20control.html>

5- PC Software control: <https://www.yahboom.net/public/upload/upload-html/1777290689/PC%20software%20control.html>

6- Dofbot-jetson-nano : https://github.com/YahboomTechnology/dofbot-jetson_nano/tree/master

7- DOFBOT SE : https://www.yahboom.net/study/DOFBOT_SE

8- DofBot manipulation : <https://www.yahboom.net/public/upload/upload-html/1699442099/4.PC%20software%20control.html>

9- Robot arm specification: <https://www.ram-e-shop.com/shop/ro-arm-dofbot-se-ai-dofbot-se-ai-vision-robotic-arm-virtual-machine-version-6dof-degrees-of-freedom-9437>

10- Dofbot manual: <https://manuals.plus/asin/B0CT7HXYCZ>

ANNEX



DOFBOT SE AI Vision Robotic Arm Virtual Machine Version 6DOF Degrees Of Freedom